

Review

Interpersonal autonomic nervous system synchrony and its association to relationship and performance – a systematic review and meta-analysis

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ABSTRACT

Interpersonal physiological synchrony is the spontaneous temporal coordination of physiological processes between several individuals. This type of synchrony is critical for human relationships, as it promotes two important outcomes: the quality of the relationships between synchronized individuals, and how well synchronized individuals perform together. Nonetheless a clear estimation of the size of the correlations between interpersonal physiological synchrony and relationship or performance outcomes is missing. To address this gap in knowledge was the main goal of the current meta-analysis. We focused on interpersonal physiological synchrony in measures of autonomic nervous system activity, and specifically we examined the distinct branches of the autonomic nervous system. We conducted two meta-analyses: (1) Estimating the association between interpersonal physiological synchrony and relationship outcomes (2) Estimating the association between interpersonal physiological synchrony and performance outcomes. In the first meta-analysis (i.e., relationships), the overall estimated correlation was small with a marginal significance ($ES=0.09$, $p>.10$) and high heterogeneity ($I^2=76.0\%$). In further sub-group analysis, we discovered a positive relationship for measures of sympathetic synchrony ($ES=0.19$, $p=.02$), a negative relationship for measures of parasympathetic synchrony ($ES=-0.21$, $p=.03$), and a positive relationship for measures of combined sympathetic and parasympathetic synchrony ($ES=0.16$, $p=.02$). As for the second meta-analysis (i.e., performance), the overall effect size of correlation was small ($ES=0.26$, $p<.01$) and heterogeneous ($I^2=52.7\%$). Our results emphasize the small effect sizes of the correlations between physiological synchrony and performance or relational outcomes, as well as the differential effects for synchrony in sympathetic versus parasympathetic activity. Combined with the high heterogeneity, our results point to the need for a rigorous a-priori scientific approach which distinguishes between different types of physiological synchrony.

1. Introduction

Synchronous interactions are a cornerstone of human experience. Widespread across various human and non-human species, synchrony has long been suggested as a type of social glue - an evolutionary-based mechanism for facilitating affiliation, bonding, cooperation, and social cohesion [1,2]. Interpersonal synchrony occurs from birth throughout life [3]. Parent-infant synchrony, namely the concordance in behaviors, emotions, and physiological processes between parent and infant, is a key aspect of social development [3]. Early synchronous experiences during development and within the parent-infant bond has also been suggested to largely influence brain development and its massive complexity [4,5]. Human bonds and social attachments are formed

through social interactions that entail rhythmic synchronous states. These are embedded within moments of asynchronous exchanges [5,6]. Even later on in life in adulthood, and even between strangers, interpersonal synchrony has emerged as a fundamental mechanism at the basis of social togetherness, cohesion and pro-sociality [7–9]. For these reasons, it is important to have a good grasp of the effects and outcomes of interpersonal physiological synchrony, that is the gap in literature we aim to fill in this current meta-analysis.

Physiological Interpersonal synchrony is defined as the temporal coordination of physiological processes between two or more individuals. It has been widely studied in recent decades [6,10–12]. The research on Interpersonal physiological synchrony has been conducted in various populations, including parent-child synchrony [13,14],

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synchrony in couples [15], synchrony during psychotherapy [16], in dyads of strangers [12] and in groups [17]. Despite this broad literature on physiological synchrony in various populations, a clear estimation of the social effects of such a synchrony and their size is yet missing. Moreover, as others have mentioned [12], the literature in this field is fragmented and hence produce non-cohesive picture. First, it contains many aspects of interpersonal physiological synchrony blended together, for example measures of the Sympathetic Nervous System (SNS) and Parasympathetic Nervous System (PNS) are usually does not given distinct consideration [18]. Second, physiological synchrony is calculated in large variety of ways (cross-correlation, spectral analysis, recurrence analysis etc.; [12,19]). Finally, physiological synchrony has been associated to various social constructs without clear theoretical distinction [12]. Hence, there is a great need for clarification and precision.

In the current study, we will focus on one type of physiological interpersonal synchrony: in measures of autonomic nervous system (ANS) activity. That is, the temporal coordination of ANS activity between two or more individuals. In this field, continuous measures of ANS activity are usually recorded from two or more individuals simultaneously, and an index of the synchrony is then calculated (as we have mentioned, in many different methodologies see [12,19]). ANS activity is usually divided into the system's two branches: Sympathetic Nervous System (SNS) and Parasympathetic Nervous System (PNS). In general, SNS is usually considered to denote increased arousal ("fight or flight") and PNS is associated with calming down and returning to regular function ("rest and digest"; [20–22]). In experiments, the functioning of these systems is usually assessed by non-invasively via the following measures: respiratory sinus arrhythmia (RSA), heart rate (HR), interbeat intervals (IBI), heart rate variability (HRV), electrodermal activity (EDA), pre-ejection period (PEP) etc. These measures relate to either the activity of SNS, PNS or both (See [23,24]; for detailed examples).

Our systematic review and meta-analysis aim to deal with several gaps in this field. First, the lack of clear estimation in effect sizes. Second, while most research in this field includes various ANS measures, it is rare to find a clear distinction between the different branches of ANS, namely SNS and PNS, hence results from SNS and PNS are many times merged. Third, different social constructs have been linked to interpersonal physiological synchrony, but the literature lacks a clear distinction between these constructs as well. Last, there are many inconsistencies related to the social meaning of physiological synchrony. At times, interpersonal synchrony was correlated with empathy, trust, and subjective togetherness [25–27], whereas at other times, it predicted negative relationship outcomes [28,54].

In order to address these gaps, we have summarized the body of evidence on the association between ANS activity, divided into its different branches, with two social constructs: relationship (within the group/dyad) and performance (of the group/dyad). Based on this evidence we present here a detailed, integrative review and meta-analysis.

2. Materials and methods

This section of the review follows guidelines detailed in the PRISMA Statement [29].

2.1. Literature search

The current review analyzed studies (published as well as unpublished studies, such as theses or dissertations) which investigate the association between interpersonal physiological synchrony (in groups or dyads) and measures of relationship or performance of the group/dyad. Since a similar systematic review of interpersonal physiological synchrony was recently published [12], reporting studies until November 2015, we have searched additional studies from 2015 until May 2020. Studies were identified by searching PsycNet and PubMed (2015-to May 2020), and complimentary search in Google Scholar has been done, to

make sure we did not miss additional studies. Additional studies were identified by examining the reference lists of target articles. Finally, we used reversed citation (i.e., "cited by") in Google Scholar for all the articles we identified. The search terms used were "interpersonal physiology", "physiological coherence", "physiological covariation", "physiological linkage" and "physiological synchrony". The search of Google Scholar, PsycNet and PubMed databases provided a total of 146 citations. 110 remained after all duplicates had been removed.

2.2. Study selection

Only English written papers were included. Studies performed on clinical population were excluded, since the aim of current review was to focus on normal population. For the same reason, studies on client-therapist physiological synchrony were excluded as well. We did not limit the population by age, in order to provide an extensive overview of the current state of results. For the same reason, we included different kinds of populations and settings, these can be divided into groups (mostly of strangers), couples and parent-child dyads.

The field of interpersonal physiological synchrony has a large number of variables and methods; hence, several initial decisions should have been made to allow systematic meta-analysis. Since we were specifically interested in the function of SNS (sympathetic nervous system) and PNS (parasympathetic nervous system), we included only measures that are known to be related to the function of these systems. For example, skin temperature could not be associated clearly to either SNS or PNS, therefore, studies including this measure were excluded. Finally, we grouped each measure to either reflecting SNS activity, PNS activity or both (see Table 1). It is worth mentioning that the interpersonal physiological synchrony was calculated and quantified in various methods, sometimes even more than one method in the same study, but for the scope of the current meta-analysis, we could not address these differences. In addition, studies that did not directly calculate physiological synchrony, but instead used a dynamical modeling (e.g. when one's physiology in the first time point predicts the other physiology in a second time point), could not be included in this meta-analysis.

Since relationship within the group/dyad and its performance are theoretically distinct concepts, we decided to a-priori conduct two different meta-analyses, one for relationship and one for group performance (somewhat similar distinction has been made by [17]). Because there was a large variety of relationship measures (see [12]), we decided to group them. Measures were considered to reflect relationship only if they clearly reflected the emotional atmosphere in the group (e.g. group satisfaction), feeling of connectedness (e.g. subjective togetherness) and understanding (e.g. empathy). Studies with measures that were only remotely related to relationship (e.g. child self-regulation) were excluded.

To sum the above, inclusion criteria were: (1) English-written papers, (2) non-clinical population, (3) the study provides a measure of interpersonal physiological synchrony, with measures that reflect SNS

Table 1
Measures associated with sympathetic nervous system (SNS) activity, parasympathetic nervous system (PNS) activity or both system's activity.

Measures associated with SNS activity	Measures associated with PNS activity	Measures associated with both SNS and PNS
Electrodermal activity (EDA; [30])	Respiratory sinus arrhythmia (RSA; [20])	Interbeat interval (IBI; [20])
Pre-ejection period (PEP; [31])	High-frequency Heart rate variability (HF – HRV; [20])	Heart rate (HR; [20])
Cardiac sympathetic index (CSI; [32])	Root mean square of the successive differences (RMSSD; [33])	Low-frequency Heart rate variability (LF – HRV; [20])
	High-frequency Pulse rate variability (HF-PRV; [75])	Low frequency Pulse rate variability (LF-PRV; [74])

activity, PNS activity or both, (4) clear measures of relationship in the group/dyad or performance must be provided in the study, (5) estimation of the association between the two above measures (3 and 4) must be provided.

Initial screening was performed on publication abstracts and final eligibility was assessed based on the full text. Data was extracted using a specially developed spreadsheet which contained detailed and standardized information: (i) type of dyad/group (couples, groups and parent-child dyads); (ii) physiological measures; (iii) outcome measures (relationship and/or performance); (iv) number of participants (or groups/dyads, in some studies) for which the study's data were analyzed (i.e. sample size); and (v) the correlation between parameters (ii) and (iii); (vi) the social context in which physiological ANS activity was recorded; (vii) the method used to quantify synchrony.

The literature search yielded 110 papers which remained after duplicates were removed. Fifty-five additional studies were included from a previous systematic review, yielding 165 studies altogether. Forty-eight papers were excluded in the initial screening step and 88 additional studies were excluded after in-depth reading. The most common causes for exclusion were a lack of relationship/performance measures, or no information on the correlation between physiological synchrony and relationship/performance measures. One study with unpublished data was added, thus, a total of 30 studies were eligible for inclusion in this review.

2.3. Data extraction

Studies in the field of interpersonal physiological synchrony usually contain many measures of relationship/performance and SNS and PNS activity. This creates several challenges in data extraction. First, many results for the same sample might cause dependency between studies. Secondly, in many studies non-significant results were not fully reported, and hence should be considered as zero correlation, which might cause underestimation of the real size effect, especially in a field with relatively small effects. To deal with these two problems, we decided: (1) From each study we extracted only one measure of the correlation between SNS, PNS or both and relationship/performance measure of the group. So, at most we had 3 different correlations from each study, and usually less. This decision allowed us to minimize dependency between studies while still including sufficient information. (2) The rule for choosing measures was the highest correlation for each system (SNS, PNS or both). Although, at surface, this might seem to inflate the power of the effects, in fact we used this rule to balance non-reported correlations (that are considered as zero correlations). Non-significant correlations that were not fully reported were considered as zero correlations. To provide an estimation of the full range of effect sizes, we also provide additional, similar tables (Tables S1 and S2), using the lowest correlation reported for each physiological system (see supplementary materials). Note that a high portion (33.33%) of the lowest correlations were not fully reported in the original papers, and hence are considered as zero for the purposes of the meta-analysis. For this reason, we could not provide a substantial meta-analysis of this data. Moreover, the fact that a high portion of the studies had unreported correlations in addition to the fact that every study provided up to three correlations, provided additional justification for our decision to systematically choose the highest correlation reported for each physiological system.

In addition, some studies reported their finding of sub-sample (e.g. men and women) separately. In this case, we included the results as different studies.

Finally, because there were both negative and positive correlations with different substantial meaning (e.g. positive correlation with marital conflict measure, which mean the synchrony has negative effect), we set the value of correlations with positive meaning (synchrony related to better relationship/performance) to be positive and the value of correlations with negative meaning (synchrony related to worse relationship/performance) to be negative.

2.4. Meta-analysis

Two meta-analyses were then planned for: (1) association between group/dyad relationship and physiological synchrony and (2) association between group/dyad performance and physiological synchrony. Relevant data were extracted and computed as Pearson correlation (in most cases, Pearson correlation was reported, and in other cases the mean, beta or chi-square were converted to Pearson correlation; (for more details see [34,35,73]).

A standard meta-analysis was performed using R statistical software (version 3.6.2), using the meta, metfor, and dmetar packages. The outcome measures for both meta-analyses were Pearson correlation between group/dyad relationship or performance and physiological synchrony. We used a random-effect model with the Sidik-Jonkman estimator [36] for the between-study heterogeneity, as our primary analysis. Between-study heterogeneity was also quantified by computing Cochran's Q test statistic and I^2 . Heterogeneity was explored using sub-groups analysis and meta-regression analysis. To investigate influential studies and outliers, we took a conservative approach that excluded only studies that were both clearly outside the range of confidence interval of overall results, and highly influential. Influence analysis was calculated by omitting one study at a time and by using Baujat Plot [37].

Possible publication bias was investigated using funnel plots (standard error of correlation estimate versus estimate of effect size for each study) and by using Egger's test of the intercept [38].

3. Results

Fig. 1 summarizes the number of records identified in each database after the duplicates and the non-English articles were removed ($N = 165$). Initial screening by title and abstract led to 48 exclusions, leaving 117 articles for full-text eligibility assessment, from which 30 were included in the meta-analysis.

3.1. Qualitative analysis

The 30 articles comprising our qualitative analysis are presented in Table 2 and Table 3. Overall, most of these articles examined groups ($N = 22$), while the minority examined couples ($N = 6$) and parent-child dyads ($N = 2$). From each article, we extracted between 1 and 3 measures, as described in Section 2.3. The first measure, regarding the association between interpersonal physiological synchrony and relationship, was reported in 26 papers with 38 results. The second measure, regarding the association between interpersonal physiological synchrony and group performance, was reported in 9 papers with 13 results.

The social context in each ANS synchrony study included in the analysis varied greatly. In the articles addressing the association between interpersonal physiological synchrony and relationship outcomes, most studies utilized paradigms in the context of group tasks ($N = 11$) or games ($N = 11$), while the minority utilized discussions ($N = 8$), non-verbal interactions ($N = 3$), video presentations ($N = 2$), lectures ($N = 1$), and romantic dates ($N = 2$). In the studies addressing interpersonal physiological synchrony and group performance, most experiments utilized group tasks paradigms ($N = 8$), while the rest were games ($N = 4$), and a discussion ($N = 1$).

The methods for quantifying synchrony were also highly heterogeneous. Most studies used methods that are based in linear cross correlations ($N = 27$) and dynamic time wrapping ($N = 7$), though there were other methods as well (with $N < 3$).

3.2. Quantitative analysis

Two meta-analyses were conducted, the first addressed the associations between interpersonal synchrony and relationship. The second

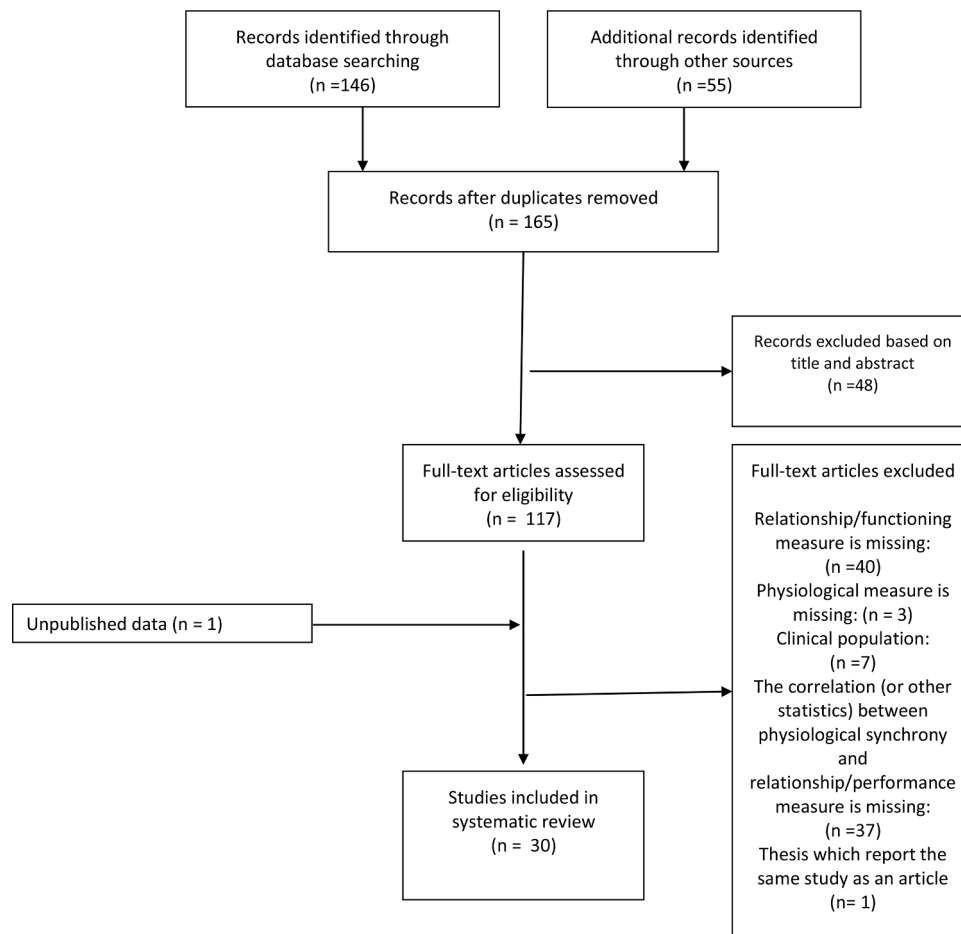


Fig. 1. A flow diagram of the studies selection process.

addressed the associations between interpersonal synchrony and performance.

3.2.1. The associations between interpersonal physiological synchrony and relationship

Twenty-six articles and 38 results met criteria for this meta-analysis. A random-effect model did not show an overall significant correlation effects (*correlation estimate*=0.07, *CI*=[−0.04, 0.18], *p*=.21; see Fig. 2) across all studies. This effect does not seem to accurately describe the data since there was considerable heterogeneity ($I^2=78.1\%$, $Q(37)=168.7$, $p<.0001$), which was expected as we included measures of both SNS and PNS in the analysis.

We then took a conservative action and looked for studies which were highly influential on overall effect and heterogeneity and also had an extreme value. Using formal influence analysis [65], no study passed the cutoff scores. Nevertheless, using The Baujat Plot [37], we found one study which contributed highly to overall heterogeneity while not having equivalent contribution to pooled effect size [58]. For this reason, we decided to exclude this study from further analyses. After deletion of the above-mentioned study, the random-effect model showed an overall estimate of correlation of 0.09 with marginal significance (*CI*=[−0.02, 0.19], *p*<.10). Nonetheless, the studies were highly heterogeneous ($I^2=76.0\%$, $Q(36)=149.98$, $p<.0001$).

Sub-group analysis with a fixed effect model, comparing effects between physiological measures associated with SNS activity, PNS activity or both, revealed a significant difference between those groups ($Q(2)=11.92$, $p<.01$). The estimate of correlation between relationship outcomes in the group and SNS activity was 0.19 ($N=14$, $p=.02$), the estimate for PNS activity was −0.21 ($N=8$, $p=.03$), and the estimate for

measures of combined SNS and PNS activity was 0.16 ($N=15$, $p=.02$). Meta-regression analysis with a mixed effect model and with the physiological system as a moderator (SNS/PNS/ SNS & PNS), revealed a significant effect ($QM(2)=10.83$, $p<.01$). The model accounted for 22.41% (R^2) of the amount of heterogeneity.

Finally, we assessed a possible publication bias. Since the overall data had high heterogeneity, which might impact the robustness of publication bias estimations [66], we performed a publication bias analysis for each sub-group separately. As for the SNS sub-group, the slight asymmetry in the funnel plot is most probably related to heterogeneity, since it is more pronounced in the negative side of the graph. As such, it is mostly caused by papers reporting negative associations between SNS activity and relationship outcomes (Fig. 3). Egger's test of the intercept [38] was not significant ($p=.25$), supporting the claim that there is no evidence for publication bias in this sub-group. The PNS sub-group's funnel plot was relatively symmetric, (Fig. 4). Egger's test of the intercept could not be performed in this case, due to the small sample size (only 8 studies). The combined SNS&PNS sub-group (Fig. 5) had a slight negative asymmetry, due to two studies, although the overall effect was positive. Egger's test of the intercept for this sub-group was also not significant ($p=.54$). Overall, the analyses of these three sub-groups did not provide evidence for publication bias in our sample.

3.2.2. The relations between interpersonal synchrony and performance

Nine articles and 13 results met criteria for meta-analysis. A random-effect model showed a marginally significant correlation effect. *Estimate*=0.21 (*CI*=[−0.00, 0.40], *p*=.06) across all studies (Fig. 6). This effect did not seem to accurately describe the data since there was considerable heterogeneity ($I^2=69.3\%$, $Q(12)=39.10$, $p<.0001$).

Table 2

The correlations between interpersonal physiological synchrony and relationship.

Study/ Year	Population	Physiological measure	Relationship measure	Sample size	Correlation	Social context	Method
[39]	Parent-child dyads	HR (SNS & PNS)	Mother-adolescent discord	32	−0.38	Plan a Fun Activity Task	Structural Equation Modeling
[40] (1)	Groups	EDA (SNS)	Positive affect	21	.37	Game playing (competitive vs. cooperative)	Linear cross-correlation
[40] (2)	Groups	HF HRV (PNS)	Social negative feeling	21	−0.34	Game playing (competitive vs. cooperative)	Weighted coherence
[40] (3)	Groups	LF HRV (SNS & PNS)	All measures	21	0 (NR)	Game playing (competitive vs. cooperative)	Weighted coherence
[41] (1)	Couples	EDA (SNS)	Empathy	14	.55	Interaction regarding memories (with/without touch)	Linear cross-correlation
[41] (2)	Couples	HF-PRV (PNS)	Empathy	14	0 (NR)	Interaction regarding memories (with/without touch)	Linear cross-correlation
[41] (3)	Couples	LF-PRV (SNS & PNS)	Empathy	14	0 (NR)	Interaction regarding memories (with/without touch)	Linear cross-correlation
[42] (1)	Groups	EDA (SNS)	Group satisfaction	53	.32	Collective intelligence tasks	Dynamic Time Warping
[42] (2)	Groups	HR (SNS & PNS)	Group satisfaction	53	0 (NR)	Collective intelligence tasks	Dynamic Time Warping
[43] (1)	Couples	EDA (SNS)	Interpersonal reactivity index for couples(females)	32	−0.33	Discussion about the positive aspects of the couple's relationship	Linear cross-correlation
[43] (2)	Couples	EDA (SNS)	Interpersonal reactivity index for couples (males)	32	.28	Discussion about the positive aspects of the couple's relationship	Linear cross-correlation
[28]	Parent-child dyads	RSA (PNS)	Children's prosocial empathy	41	−0.37	Watching a film with negative valence together	Linear cross-correlation
[44]	Groups	PEP (SNS)	Positive affect	106	.31	Playing a game	Linear cross-correlation
[45] (1)	Groups	CSI (SNS)	Perceived similarity	134	.12	Cooperative/competitive task	Multivariate modeling
[45] (2)	Groups	LogRSA (PNS)	Perceived Similarity	134	−0.05	Cooperative/competitive task	Multivariate modeling
[46]	Groups	HR (SNS & PNS)	Relatedness	29	.14	Lego construction tasks	Cross recurrence quantification analysis
[47]	Groups	EDA (SNS)	Affective valance	24	.32	Lecture	Dynamic time warping
[48] (1)	Couples (husbands)	RSA (PNS)	Marital conflict	49	−0.31	Tetradic family free-play	Linear cross-correlation
[48] (2)	Couples (wives)	RSA (PNS)	Marital conflict	49	−0.35	Tetradic family free-play	Linear cross-correlation
[49]	Groups	IBI (SNS & PNS)	Commitment	150	.18	Group discussion	Linear cross-correlation
[50]	Groups	IBI (SNS & PNS)	Group cohesion	47	.34	Drumming task	Linear cross-correlation
[51]	Groups	HR (SNS & PNS)	Comprehension	57	.29	Watching video clips	Linear cross-correlation
[25] (1)	Groups	EDA (SNS)	All measures	82	0 (NR)	Computer game	Linear cross-correlation
[25] (2)	Groups	HF HRV (PNS)	Perceived comprehension	82	.26	Computer game	Linear cross-correlation
[25] (3)	Groups	IBI (SNS & PNS)	Emotional empathy	82	.34	Computer game	Linear cross-correlation
[52] (1)	Groups	EDA (SNS)	Group satisfaction	28	.52	Solving problems	Dynamic Time Warping
[52] (2)	Groups	HR (SNS & PNS)	Group satisfaction	28	.49	Solving problems	Dynamic Time Warping
[53]	Groups	EDA (SNS)	Liking	12	−0.36	Discussions	Spearman's rank-order coefficient
[54]	Couples	Physiological Index (SNS & PNS)	Marital satisfaction	30	−0.31	Discussing marital problems	Linear cross-correlation
[26]	Groups	HR (SNS & PNS)	Trust	57	.43	Lego building	Multivariate Recurrence Quantification Analysis
[55]	Groups	EDA (SNS)	Tensions within the group	153	−0.18	Constructing origami boats	Cross-Recurrence Quantification Analysis
[27]	Groups	HR (SNS & PNS)	Subjective togetherness	9	.38	Playing the mirror game	Pearson correlation
[56] (1)	Couples	EDA (SNS)	Attraction	140	.17	Dating	Linear cross-correlation
[56] (2)	Couples	IBI (SNS & PNS)	Attraction	140	.14	Dating	Linear cross-correlation
[57]	Groups	EDA (SNS)	Collaboration (reciprocal interaction)	39	.47	Computational task	Directional agreement
[58]	Groups	IBI (SNS & PNS)	Trust	40	−0.56	Video game	Cross-fuzzy entropy
[59]	Groups	IBI (SNS & PNS)	Perceived empathy	70	.18	discussion tasks with/without feedback about physiological synchrony	Linear cross-correlation
[60]	Couples	HF-HRV (PNS)	Momentary closeness	86	−0.46	Marital problem discussion	Linear cross-correlation

Note. EDA = Electrodermal activity; RSA = Respiratory sinus arrhythmia; HF-HRV = High-frequency heart rate variability; IBI = Interbeat interval; HR = Heart rate; PEP = Pre-ejection period; LF-HRV = Low-frequency heart rate variability; CSI=Cardiac sympathetic index; PRV= Pulse rate variability; SNS = Sympathetic nervous system; PNS = Parasympathetic nervous system; NR= Not fully reported in the study. The numbers in the brackets denote different measures from the same study. The sample size is based on the analyses performed in each study; At times, the analysis was at the individual-level, and at times it was made at the group/dyad-level.

Comparable to the previous section, we looked for studies that are both highly influential and with extreme values. A formal influence analysis [65] revealed that no study passed the cutoff scores. Using The Baujat Plot [37], we found one study that contributed greatly to overall heterogeneity. Moreover, this study had an extreme value, in relation to the overall estimate, hence we deleted it from our analysis [58]. After deletion of the above-mentioned study, a random-effect model revealed an overall estimated correlation of $r = 0.26$ ($CI=[.07, 0.42]$, $p<.01$)

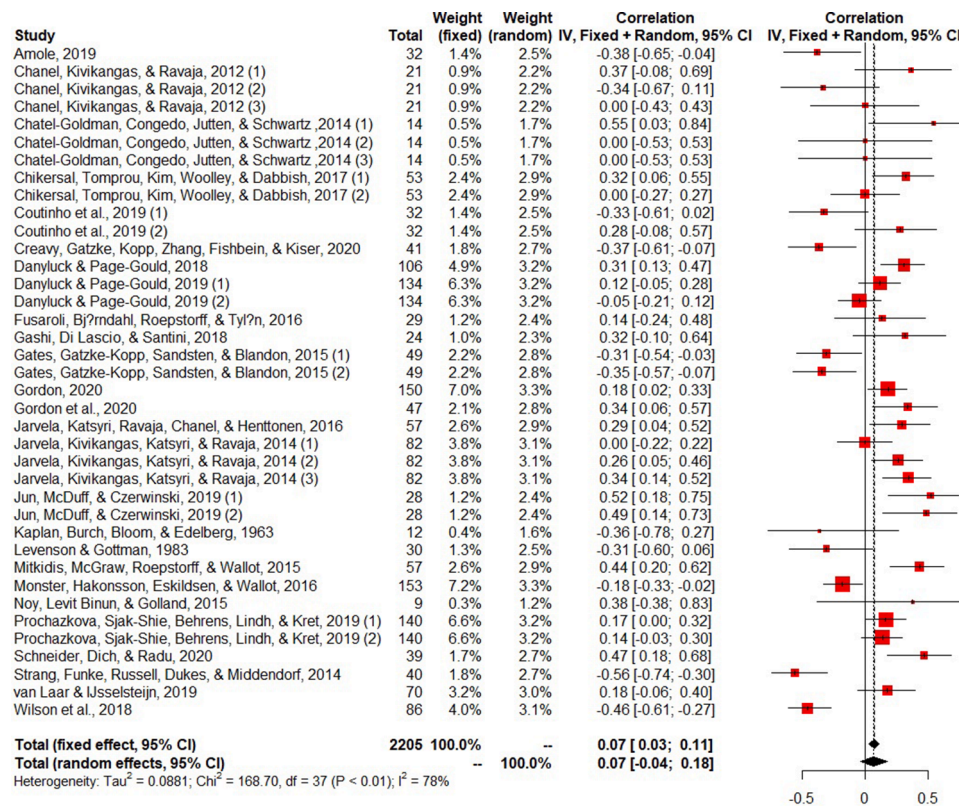
across all studies. We found high level of heterogeneity ($I^2=52.7\%$, $Q(11)=23.24$, $p<.02$), but due to the small sample, further sub-groups analyses could not be conducted. Next, we assessed publication bias using a funnel plot, which revealed some a-symmetry towards the positive side of the graph (Fig. 7). Egger's test of the intercept [38] was significant, ($p<.01$) raising the possibility for publication bias. To deal with the possible publication bias, we used the sample size as a moderator (see, for example, [67]), to explore whether it influenced

Table 3

The correlations between interpersonal physiological synchrony and group performance.

Study/ Year	Population	Physiological measure	Performance measure	Sample size	Correlation	Social context	Method
[42] (1)	Groups	HR (SNS & PNS)	Collective Intelligence	53	−0.14	Collective intelligence tasks	Dynamic Time Warping
[42] (2)	Groups	EDA (SNS)	Collective Intelligence	53	−0.064	Collective intelligence tasks	Dynamic Time Warping
[61] (1)	Groups	logRSA (PNS)	Performance	10	.63	Military task	Pearson correlation
[61] (2)	Groups	IBI (SNS & PNS)	Performance	10	.65	Military task	Directional agreement
[50]	Groups	IBI (SNS & PNS)	Performance	47	.32	Drumming task	Linear cross- correlation
[62] (1)	Groups	EDA (SNS)	Task completion time	16	.28	Computer game	Weighted coherence
[62] (2)	Groups	IBI (SNS & PNS)	Task completion time	16	.54	Computer game	Linear cross- correlation
[63]	Groups	IBI (SNS & PNS)	Tracking error	16	.65	Computer game	Linear cross- correlation
[64] (1)	Groups	EDA (SNS)	Performance	24	.33	Computerized task	Directional agreement
[64] (2)	Groups	IBI (SNS & PNS)	Performance	24	0 (NR)	Computerized task	All methods
[57]	Groups	EDA (SNS)	Performance	39	.28	Collaborative learning	Signal matching
[58]	Groups	IBI (SNS & PNS)	Performance	40	−0.47	Video game	Cross-fuzzy entropy
[59]	Groups	IBI (SNS & PNS)	Performance	70	.1	Discussion tasks with/without feedback about physiological synchrony	Linear cross- correlation

Note. EDA = Electrodermal activity; RSA = Respiratory sinus arrhythmia; IBI = Interbeat interval; HR = Heart rate; SNS = Sympathetic nervous system; PNS = Parasympathetic nervous system; NR = Not fully reported in the study. The numbers in the brackets denote different measures from the same study. The sample size is based on the analyses made in each study, at times the analysis was made on individuals, and at times it was made in group/dyad level.

**Fig. 2.** A forest plot of meta-analysis of the associations between interpersonal physiological synchrony and relationship outcomes.

effect sizes. Indeed, a meta-regression analysis with a mixed effects model and with sample size as a moderator, revealed a significant effect to sample size ($QM(2) = 6.18$, $p = .01$). Effect size was negatively related to sample size ($estimate = -0.01$, $p = .01$), which means that smaller studies reported larger effects. The overall estimate in this model was not significant ($estimate = 0.12$, $CI = [-0.10, 0.34]$, $p = .28$).

4. Discussion

This comprehensive review and meta-analyses were conducted in order to explore the associations between (1) interpersonal physiological synchrony of ANS activity and relationship outcomes in groups or dyads and (2) interpersonal physiological synchrony of ANS activity and

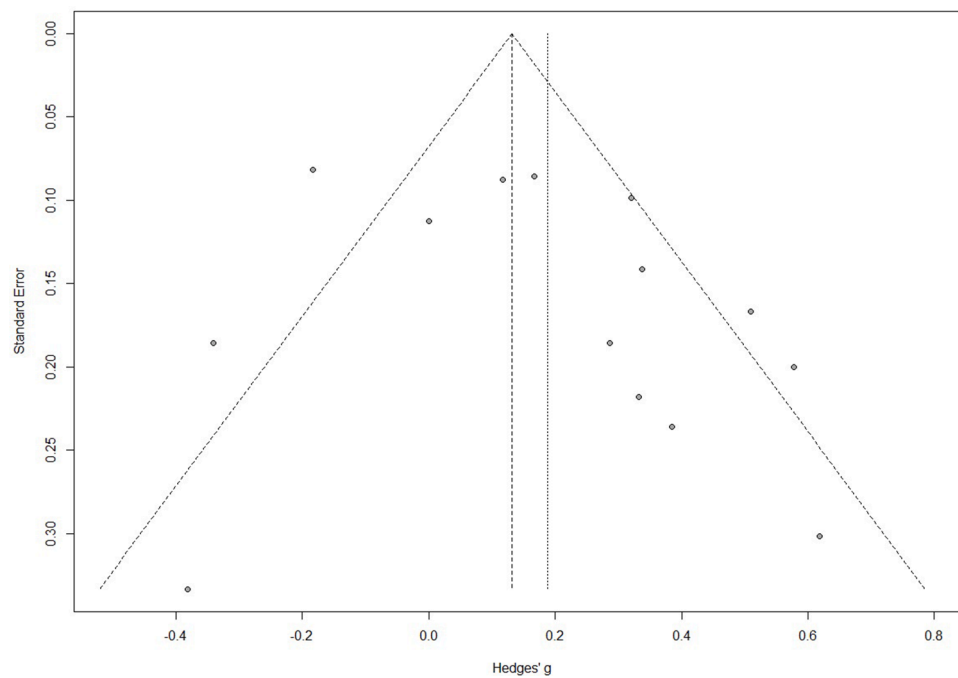


Fig. 3. A funnel plot of the associations between interpersonal sympathetic nervous system synchrony and relationship outcomes. Standard error of the correlation estimates versus estimates of effect size for each study are presented in the vertical axis.

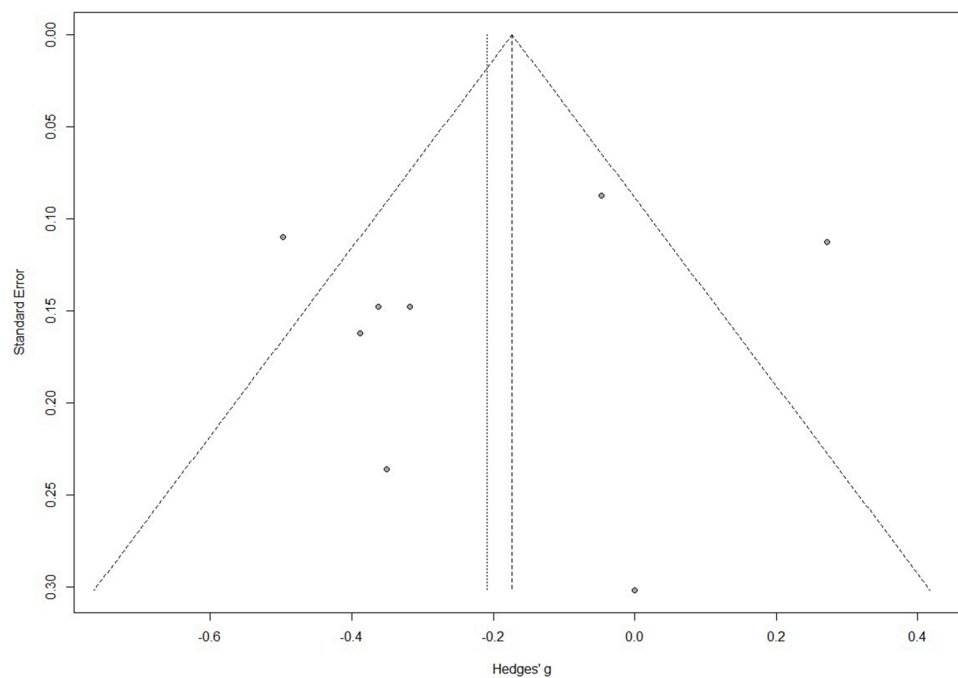


Fig. 4. A funnel plot of the associations between interpersonal parasympathetic nervous system synchrony and relationship outcomes. Standard errors of the correlation estimates versus the estimates of effect size for each study are presented in the vertical axis.

performance outcomes in groups or dyads.

In the first meta-analysis, we found a small effect size, with marginal significance. Nonetheless, it was clear that this overall estimation did not accurately represent the data since it was highly heterogenous. Indeed, sub-group analysis and meta-regression analysis revealed differences between interpersonal synchrony in measures that relate to SNS, PNS or SNS and PNS combined. For SNS activity, interpersonal synchrony was positively correlated ($r = 0.19$) to relationship outcomes. Namely, enhanced SNS synchrony was related to better relationship

outcomes. On the other hand, synchrony in PNS activity was negatively related ($r = -0.21$) to relationship in the group. Namely, enhanced synchrony in measures of the PNS was negatively related to better relationship outcomes. So, better relationship in the group was related to more (in-phase) SNS synchrony and less (in-phase) PNS synchrony. Finally, measures that reflected increased synchrony in both SNS and PNS activity, were related to better relationship, but with smaller effect ($r = 0.16$).

It was previously suggested [12] that interpersonal physiological

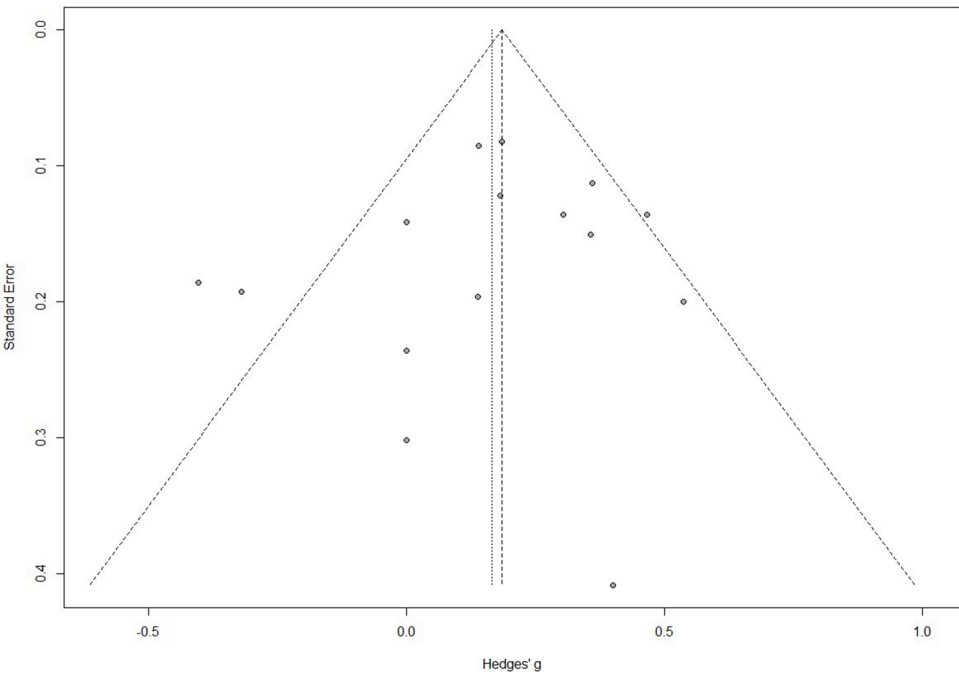


Fig. 5. A funnel plot of the associations between interpersonal sympathetic & parasympathetic nervous system synchrony and relationship outcomes. Standard errors of the correlation estimates versus the estimates of effect size for each study are presented in the vertical axis.

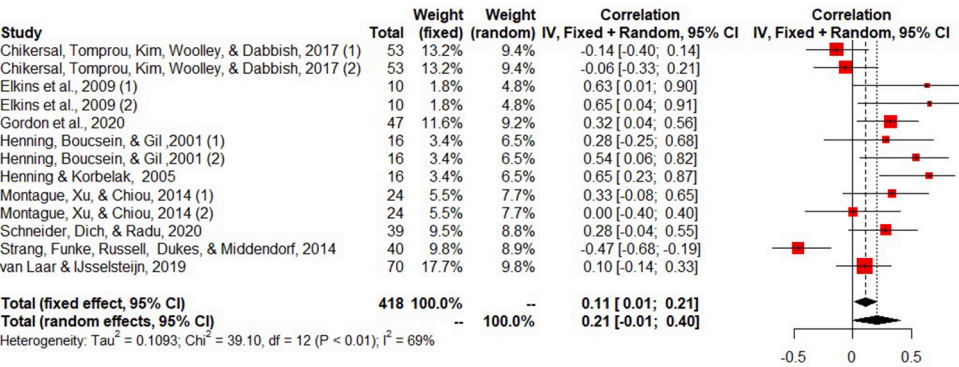


Fig. 6. A forest plot of the meta-analysis of the associations between interpersonal physiological synchrony and performance outcomes.

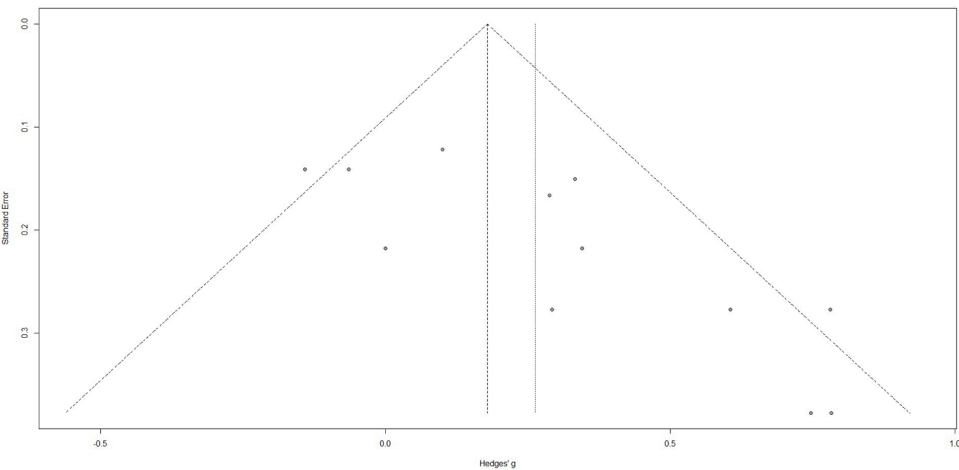


Fig. 7. A funnel plot of the associations between interpersonal sympathetic & parasympathetic nervous system synchrony and performance outcomes. Standard errors of the correlation estimates versus the estimates of effect size for each study are presented on the vertical axis.

synchrony reflects emotional empathy. That is, one's ANS activity, which might reflect her or his arousal and overall emotional state, is mirrored to another's ANS activity and emotional state. According to this explanation, interpersonal ANS synchrony is a state that occurs when people are feeling the same, and thus understand each other from within. Feldman [4] considered ANS synchrony as a developmental process, emerging from birth, between mother and her infant. ANS synchrony is thus considered a part of a bio-behavioral matrix, a cornerstone of bonding and affiliation, which encompasses various processes such as attachment and co-regulation of emotions and neurobiological states. Other authors [68] have also emphasized that ANS synchrony is a co-regulatory process.

In contrast, there is literature suggesting that anti-phase ANS synchrony (that is, the negative association in ANS activity) might be adaptive as well. For example, Levenson & Gottman [54] suggested that in-phase ANS synchrony in couples reflects being "locked into" the interaction, so when one partner is feeling bad, the other partner is unable to step back, and regulate the other partner. In their view, ANS synchrony at times reflects a state of rigidity and is thus associated with decreased relationship quality. Gates and colleagues ([48]) suggest that this negative association of ANS activity between individuals might reflect a dynamic of co-regulatory homeostasis, in other words, the tendency to reduce one's arousal when the other's arousal is increased.

These seemingly contradictory predications about the association between ANS synchrony and relationship, fall in line with our results. While SNS positive (in-phase) synchrony was related to better relationship outcomes, PNS in-phase synchrony was related to poorer relationship outcomes. An understanding of the difference in SNS, PNS, their combined functioning and their social meaning may be critical in taking the research forward and providing fuller explanations for results. SNS activity is mainly a sign of arousal, thus mutual change in arousal might reflect being with each other in the same emotional state. The other side of the coin might be, that when SNS activity is not coordinated, individuals feel less connected to each other, hence their relationship is conceived as poor. Conversely, PNS activity is related to different functions. According to the polyvagal theory ([69], 2011), increased PNS activity in non-threatening situations is related to social engagement and emotional regulation. That is, heightened PNS activity occurs when individuals engage with each other in a social environment that is perceived as safe. In turn, low PNS activity might be a sign of a failure to regulate emotion and being in a "fight or flight" situation in which the SNS takes over regulation. Taking that into account, a concurrent mutual low PNS activity may reflect situations where both individuals feel unsafe and refrain from social engagement. So, we suggest that the negative association between PNS synchrony and relationship potentially indicates a context in which both individuals were failing to regulate themselves and each other. The smaller effect for measures that related to both systems' functioning, may imply that in the context of relationships, the sympathetic and parasympathetic activity has an additive component, that reflects the positive association with sympathetic (in-phase) synchrony and the negative association with parasympathetic (in-phase) synchrony (for further discussion of the balance between these systems, see [70]).

In our second meta-analysis, assessing the association between ANS synchrony and group performance, we also found a small effect size ($r = 0.26$), although the data was highly heterogenous. Since there were relatively few studies, we could not conduct a subgroup and meta regression analysis. This finding is in line with most research, that found small associations between ANS synchrony and performance of the group although the results are somewhat mixed [17]. Nonetheless, our meta-analysis is the first to provide an estimation of the size for this effect.

Our qualitative analysis regarding the methods for quantifying synchrony revealed that most studies used methods that are based on linear cross-correlation (with or without lag). Nevertheless, many other methods exist in the field, from which the most common is dynamic time

wrapping. Although we could not conduct meta-analysis on this subject, the large variety of methods calls for better definitions of the different methods and their theoretical meaning. There is also great need for rigorous comparison between these methods. The social context of the studies was also very diverse, a fact that was probably contributes highly to overall heterogeneity of effect sizes.

4.1. Limitations

Our systematic review and meta-analyses have several limitations. First, we deliberately included only non-clinical populations which limits the generalizability of our conclusions to neurotypical populations only. On the other hand, since our populations varied greatly, and included romantic couples, groups of strangers and mother-child dyads, it was beyond the scope of our aims to include clinical samples. Further investigations should systematically explore similar results from clinical samples and compare them to the current results. Second, since the methods of calculating physiological synchrony vary greatly between different reports, we could not consider the differences between these methods. However, as Butler [71] has mentioned, different methods may imply different theoretical meaning (e.g. morphostatic vs. morphogenic processes) and we are unable to interpret such differences in the current analysis. Third, the studies in this field rarely contain only a single measure per physiological system, hence there is danger of both false positive results and false negative errors. Our data extraction took into consideration both these errors, but still, they may affect our overall results. This limitation might also explain, at least partly, the overall high heterogeneity in the analyses.

We found some evidence for a publication bias, showing that small studies produced higher correlations, and that the overall estimate of the associations between ANS activity and performance were non-significant. This raises the possibility of publication bias in the field, although the relatively small number of studies and high heterogeneity does not allow to draw firm conclusions.

4.2. Future recommendation

First and foremost, our systematic review and meta-analyses points at a great need to improve scientific standards in the field. The term physiological synchrony describes various phenomenon in the literature. Following others [71, 19], we further emphasize the need to be clearer and to use different terminology with a distinct meaning when studying physiological synchrony. Among them, we can highlight previous suggestions made regarding the distinction between in-phase and anti-phase synchrony [71, 72], between morphostatic and morphogenic processes [71], and between methods of calculation that are based on time and those that based on frequencies [19]. The term synchrony alone does not capture all these characteristics, but these characteristics may determine the meaning of synchrony and the expected direction of results. Most specifically, our meta-analysis emphasizes the importance of the distinction between SNS, PNS, and their combined synchrony, which are usually grouped together under the term ANS synchrony. We call for enhanced a-priori rigor in the field that takes into account these meaningful distinctions. Furthermore, the high heterogeneity might imply that an overall estimation of synchrony does not capture important parts which are the essence of interpersonal physiology and there is a need for an even more detailed description of the interpersonal physiological dynamics [6]. Lastly, as others have mentioned [12] the social context is crucial in understanding physiological interpersonal synchrony, for example synchrony in a negative social situation can potentially be related to different physiological dynamics than synchrony in a positive social situation. These important influences of context need to be incorporated in future research designs.

5. Conclusions

The association between general ANS interpersonal synchrony and relationship outcomes in the group/dyad had a small effect size, and so did the association between general ANS interpersonal synchrony and groups' performance outcomes. The first association was also dependent on the specific physiological system: SNS synchrony tended to be positively correlated to relationship outcomes in the group, while PNS synchrony was negatively associated with relationship in the group. Synchrony in measures of SNS and PNS combined were also positively associated with relationship outcomes, but with even a smaller effect size. Our investigation points at various pitfalls in this field and calls for improving scientific standard and theoretical precision in the physiological interpersonal synchrony research field.

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Declaration of Competing Interest

Authors declare no conflict of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.physbeh.2021.113391](https://doi.org/10.1016/j.physbeh.2021.113391).

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